Lecture 6: Trigonometric Integrals

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Based on Thomas' Calculus, 15th Edition in SI Units, Pearson Education Limited

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Integrating powers of sines and cosines.
- ☐ Integrating powers of $\tan x$ and $\sec x$.
- ☐ Integrating multiple angles of sine and cosine.

Introduction

Trigonometric integrals involve algebraic combinations of the six basic trigonometric functions.

The general idea is to use the identities to 'transform' the integrals into the format that are easier to work with.

Integrating powers of sine and cosine

Integrals of the form $\int \sin^m x \cos^n x \, dx$ where m and n are nonnegative integers.

There are essentially two cases:
(a) either m or n (or both) is odd;
(b) both m and n are even.

Case 1: Either m or n (or both) is odd

- When m is odd and n is even
- When m is even and n is odd
- When both m and n are odd

Integrating powers of sine and cosine

$$\int \sin^3 x \cos^2 x \, dx$$

$$\int \cos^5 x \, dx$$

Integrating powers of sine and cosine

$$\int \sin^3 x \cos^5 x \, dx$$

Integrating powers of sine and cosine

Case 2: Both m and n are even

$$\int \sin^2 x \cos^4 x \, dx$$

Eliminating Square Roots

$$\int_0^{\pi/4} \sqrt{1+\cos 4x}\, dx$$

Integrating powers of $\tan x$ and $\sec x$

Use the identity

$$\tan^2 x + 1 = \sec^2 x$$

to integrate by substitution or by parts when necessary.

$$\int \tan^4 x \, dx$$

Integrating powers of $\tan x$ and $\sec x$

$$\int \sec^3 x \, dx$$

$$\int \tan^4 x \sec^4 x \, dx$$

Integrating multiple angles of sine and cosine

Integrals of the form

$$\int \sin mx \sin nx \, dx, \quad \int \sin mx \cos nx \, dx, \quad \text{and} \quad \int \cos mx \cos nx \, dx.$$

Use the identities

$$\sin mx \sin nx = \frac{1}{2}[\cos(m-n)x - \cos(m+n)x],$$

$$\sin mx \cos nx = \frac{1}{2}[\sin(m-n)x + \sin(m+n)x],$$

$$\cos mx \cos nx = \frac{1}{2}[\cos(m-n)x + \cos(m+n)x].$$

Integrating multiple angles of sine and cosine

$$\int \sin 3x \cos 5x \, dx$$

$$\int \cos 5x \cos 3x \, dx$$

Integrating multiple angles of sine and cosine

$$\int \sin 3x \sin 2x \, dx$$